

CENTRAL BOARD OF SECONDARY EDUCATION
HALF-YEARLY PRACTICE EXAMINATION (SESSION 2024-2025)
SUBJECT: SCIENCE (CLASS - IX) | CHAPTER 8: JOURNEY INSIDE THE ATOM

TIME ALLOWED: 45 MINUTES**MAXIMUM MARKS: 25****PAPER CODE: 086/CH08****Student's Name:** _____**Roll No.:** _____**Class & Section:** Class IX - _____**Date of Exam:** _____**Invigilator's Sign:** _____**GENERAL INSTRUCTIONS:**

1. The question paper comprises **25 Multiple Choice Questions (MCQs)** of **1 mark each**.
2. **Section A (Q1 – Q15):** Standard Multiple Choice Questions with one correct option.
3. **Section B (Q16 – Q20):** Assertion-Reasoning based questions. Read both statements carefully before marking.
4. **Section C (Q21 – Q25):** Case-Study / Practical Data based questions.
5. All questions are compulsory. There is no negative marking.
6. Darken the corresponding circle on the **OMR Sheet** using a blue or black ballpoint pen.

SECTION A: MULTIPLE CHOICE QUESTIONS (Q.1 TO Q.15)**[15 MARKS]****Q.1 Who discovered the cathode rays (negatively charged subatomic electrons) in 1897?**

- | | |
|-----------------------|--------------------|
| (A) John Dalton | (B) J.J. Thomson |
| (C) Ernest Rutherford | (D) James Chadwick |

Q.2 Canal rays discovered by E. Goldstein in 1886 led to the identification of which positively charged subatomic particle?

- | | |
|--------------|--------------|
| (A) Electron | (B) Proton |
| (C) Neutron | (D) Positron |

Q.3 In Rutherford's famous alpha-particle scattering experiment, what metal foil was used because of its exceptional malleability?

- | | |
|-------------------|-------------------|
| (A) Aluminum foil | (B) Gold foil |
| (C) Silver foil | (D) Platinum foil |

Q.4 What surprising observation in Rutherford's alpha scattering experiment led to the discovery of the atomic nucleus?

- | | |
|---|--|
| (A) All alpha particles were absorbed | (B) Most particles were deflected by 90° |
| (C) Nearly 1 in 12,000 alpha particles rebounded backwards at nearly 180° | (D) Alpha particles turned into beta particles |

Q.5 What major drawback of Rutherford's nuclear model was successfully resolved by Neils Bohr's atomic model?

- | | |
|---|--|
| (A) Failure to explain mass of neutron | (B) The orbital instability of accelerating electrons radiating energy and collapsing into the nucleus |
| (C) Inability to predict chemical bonds | (D) Inability to explain atomic weight of chlorine |

Q.6 According to the Bohr-Bury scheme, what is the maximum number of electrons that can be accommodated in the $5M$ -shell ($5n = 35$)?

- | | |
|--------|--------|
| (A) 8 | (B) 18 |
| (C) 32 | (D) 2 |

Q.7 An atom has atomic number $Z = 16$. What is the electron distribution in its K, L, and M shells?

- (A) 2, 8, 6 (B) 2, 6, 8
(C) 8, 6, 2 (D) 2, 14

Q.8 What is the valency of a chlorine atom having atomic number 17 and electronic configuration 2, 8, 7?

- (A) 7 (B) 1
(C) 8 (D) 3

Q.9 Who discovered the neutral subatomic particle 'neutron' in 1932?

- (A) Ernest Rutherford (B) James Chadwick
(C) Niels Bohr (D) Henry Moseley

Q.10 An atom of an element has 11 protons, 12 neutrons, and 11 electrons. What are its atomic number (Z) and mass number (A)?

- (A) $Z = 11$, $A = 12$ (B) $Z = 11$, $A = 23$
(C) $Z = 12$, $A = 23$ (D) $Z = 23$, $A = 11$

Q.11 Atoms of the same element having the same atomic number but different mass numbers are termed:

- (A) Isobars (B) Isotopes
(C) Isotones (D) Isomers

Q.12 Which radioactive isotope is widely utilized in medical therapy for the treatment of cancer?

- (A) Carbon-14 (B) Cobalt-60
(C) Iodine-131 (D) Uranium-235

Q.13 Naturally occurring chlorine exists as two isotopes: $^{35}_{17}\text{Cl}$ (75%) and $^{37}_{17}\text{Cl}$ (25%). The average atomic mass of chlorine is:

- (A) 35.0 u (B) 36.0 u
(C) 35.5 u (D) 37.0 u

Q.14 Calcium ($^{40}_{20}\text{Ca}$) and Argon ($^{40}_{18}\text{Ar}$) have different atomic numbers but the same mass number (40). They are termed:

- (A) Isotopes (B) Isobars
(C) Allotropes (D) Homologues

Q.15 What is the maximum number of electrons that can be accommodated in the outermost valence shell of any stable atom?

- (A) 2 (B) 8
(C) 18 (D) 32

SECTION B: ASSERTION - REASON QUESTIONS (Q.16 TO Q.20)

[5 MARKS]

Directions for Q16 to Q20: Select the correct option from the choices given below:

- (A) Both Assertion (A) and Reason (R) are true and Reason (R) is the correct explanation of Assertion (A).
(B) Both Assertion (A) and Reason (R) are true but Reason (R) is NOT the correct explanation of Assertion (A).
(C) Assertion (A) is true but Reason (R) is false.
(D) Assertion (A) is false but Reason (R) is true.

Q.16 Assertion (A): An atom as a whole is electrically neutral.

Reason (R): The number of positively charged protons in the nucleus equals the number of negatively charged electrons revolving around it.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.17 Assertion (A): Isotopes of an element have identical chemical properties.

Reason (R): Chemical properties depend on the number and arrangement of valence electrons, which is identical in all isotopes of an element.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.18 Assertion (A): The mass of an atom is almost entirely concentrated inside its central nucleus.

Reason (R): Protons and neutrons (nucleons) reside in the nucleus, and the mass of electrons is negligibly small ($\frac{1}{1840}$ of a proton).

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.19 Assertion (A): Noble gases like Helium, Neon, and Argon exhibit zero valency.

Reason (R): Their outermost shells have completely filled octets (or duplet in Helium), rendering them chemically unreactive.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

Q.20 Assertion (A): Thomson's atomic model is known as the plum pudding model.

Reason (R): Thomson proved that all the positive charge is concentrated in a tiny central nucleus.

- (A) Both (A) and (R) are true and (R) is the correct explanation of (A).
 (B) Both (A) and (R) are true but (R) is NOT the correct explanation of (A).
 (C) (A) is true but (R) is false.
 (D) (A) is false but (R) is true.

SECTION C: CASE-BASED / EXPERIMENTAL QUESTIONS (Q.21 TO Q.25)

[5 MARKS]

READING PASSAGE & EXPERIMENTAL SETUP:

Case Study: Atomic Structure of Elements X, Y, and Z

Study the table below representing three neutral elements and answer questions 21 to 25:

Element	Atomic Number (Z)	Mass Number (A)	Neutrons
Element X	6	12	6
Element Y	6	14	8
Element Z	7	14	7

Q.21 What is the relationship between Element X and Element Y?

- (A) They are isobars (B) They are isotopes of carbon
(C) They are allotropes of nitrogen (D) They have different numbers of electrons

Q.22 What is the relationship between Element Y and Element Z?

- (A) They are isotopes (B) They are isobars
(C) They belong to the same group (D) They have identical chemical properties

Q.23 What is the electronic configuration of Element X?

- (A) 2, 4 (B) 2, 8, 2
(C) 6, 0 (D) 2, 2, 2

Q.24 What is the valency of Element Z (atomic number 7)?

- (A) 5 (B) 3
(C) 7 (D) 1

Q.25 Which radioactive isotope among X, Y, or Z is used in archaeological radiocarbon dating to determine the age of ancient fossils?

- (A) Element X (Carbon-12) (B) Element Y (Carbon-14)
(C) Element Z (Nitrogen-14) (D) None of these

CBSE CANDIDATE OMR ANSWER RESPONSE GRID

01 (A) (B) (C) (D)	02 (A) (B) (C) (D)	03 (A) (B) (C) (D)	04 (A) (B) (C) (D)	05 (A) (B) (C) (D)
06 (A) (B) (C) (D)	07 (A) (B) (C) (D)	08 (A) (B) (C) (D)	09 (A) (B) (C) (D)	10 (A) (B) (C) (D)
11 (A) (B) (C) (D)	12 (A) (B) (C) (D)	13 (A) (B) (C) (D)	14 (A) (B) (C) (D)	15 (A) (B) (C) (D)
16 (A) (B) (C) (D)	17 (A) (B) (C) (D)	18 (A) (B) (C) (D)	19 (A) (B) (C) (D)	20 (A) (B) (C) (D)
21 (A) (B) (C) (D)	22 (A) (B) (C) (D)	23 (A) (B) (C) (D)	24 (A) (B) (C) (D)	25 (A) (B) (C) (D)

Total Questions Attempted: ____ / 25

Marks Obtained: ____ / 25

Evaluator's Signature: _____

OFFICIAL ANSWER KEY & DETAILED EXPLANATIONS

Q. No.	Correct	Conceptual Explanation & Syllabus Rationale
Q.1	(B)	J.J. Thomson discovered the electron through cathode ray tube experiments, demonstrating that atoms are divisible.
Q.2	(B)	E. Goldstein observed positively charged radiation passing through perforations in cathode rays (canal rays), identifying protons.
Q.3	(B)	Rutherford used gold foil because it could be hammered extremely thin (about 1000 atoms thick).
Q.4	(C)	Alpha particles rebounding straight back showed that positive charge and almost all mass are concentrated in a tiny dense core (nucleus).
Q.5	(B)	Classical physics stated accelerating charges radiate energy. Bohr proposed discrete non-radiating stationary energy orbits.
Q.6	(B)	Maximum electrons $= 2n^2$. For M-shell ($n=3$): $2(3^2) = 2 \times 9 = 18$.
Q.7	(A)	Sulfur ($Z=16$): K-shell holds 2, L-shell holds 8, remaining 6 enter M-shell (2, 8, 6).
Q.8	(B)	Chlorine needs 1 electron to complete its valence octet. Valency $= 8 - 7 = 1$.
Q.9	(B)	James Chadwick discovered neutrons by bombarding beryllium with alpha particles, observing neutral radiation of mass ≈ 1 u.
Q.10	(B)	$Z = \text{protons} = 11$. $A = \text{protons} + \text{neutrons} = 11 + 12 = 23$ (Sodium).
Q.11	(B)	Isotopes have identical atomic numbers (same chemical properties) but different neutron counts (different mass numbers).
Q.12	(B)	Cobalt-60 emits penetrating gamma rays utilized in targeted radiation therapy for cancer.
Q.13	(C)	$\text{Average mass} = \left(35 \times \frac{75}{100}\right) + \left(37 \times \frac{25}{100}\right) = 26.25 + 9.25 = 35.5$ u.
Q.14	(B)	Isobars are atoms of different elements having different atomic numbers but identical mass numbers.
Q.15	(B)	According to the octet rule of the Bohr-Bury scheme, the outermost valence shell cannot accommodate more than 8 electrons.
Q.16	(A)	Because proton charge ($+1$) balances electron charge (-1) and their numbers are equal, the net atomic charge is zero.
Q.17	(A)	Chemical bonding involves valence electrons. Since isotopes have identical electron configurations, their chemical behavior is identical.
Q.18	(A)	Each proton and neutron has a mass of ≈ 1 u, while an electron has negligible mass (0.00054 u).
Q.19	(A)	With complete valence shells, noble gases do not gain, lose, or share electrons, hence valency $= 0$.
Q.20	(C)	Assertion is true. Reason is false: Thomson proposed positive charge is distributed uniformly throughout a sphere; the tiny nucleus was discovered by Rutherford.
Q.21	(B)	Both have $Z = 6$ (carbon) but different mass numbers (12 and 14); hence, they are isotopes.
Q.22	(B)	Element Y (${}^{14}_6\text{C}$) and Element Z (${}^{14}_7\text{N}$) have different atomic numbers (6 and 7) but the same mass number (14); they are isobars.
Q.23	(A)	For $Z = 6$, electrons $= 6$. Distribution is K-shell = 2, L-shell = 4 (2, 4).
Q.24	(B)	For $Z = 7$, configuration is 2, 5. To complete its octet, it requires 3 electrons. Valency $= 8 - 5 = 3$.
Q.25	(B)	Carbon-14 (Element Y) is a beta-emitting radioactive isotope with a half-life of 5,730 years used in radiocarbon dating.